4.4 – Congruence and Transformations • Homework

Name: _____

Name the single rigid motion that maps the preimage onto the image.

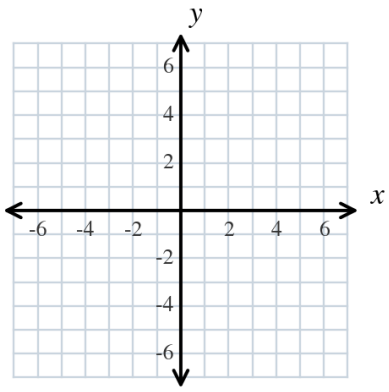
1. A(−3, 5) → A′(3, 5)

2. B(−3, 6) → B′(2, 4)

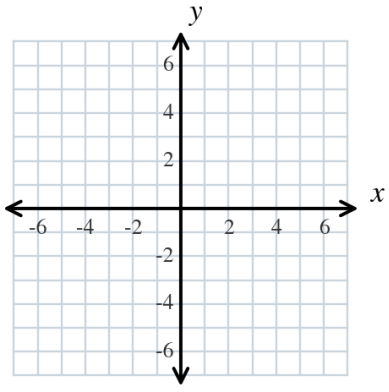
3. C(4, −7) → C′(−4, 7)

Find the final image after each composition.

4. D(2, −5) after a reflection in the x-axis, then the translation (x, y) → (x, y + 8).



5. △EFG with E(1, 1), F(4, 2), and G(3, −2) after a 90° rotation about the origin, then a reflection in the y-axis. Graph both.



Use the theorems about reflections in two lines.

6. Lines m: y = 2 and n: y = 5 are parallel. Reflect in m, then in n. Write ONE rule for the composition, and check it with (4, 1).

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7. Reflect in the x-axis, then in the y-axis. What single transformation is this? Check it with (3, 1).

Answer each question.

8. A figure is reflected in two parallel lines 7 units apart. How far does the final image sit from the preimage, and in which direction?

9. Jo says a composition of three reflections cannot be a rigid motion because there are too many steps. Is Jo right? Explain.

10. $\triangle ABC \cong \triangle XYZ$. Does that guarantee a congruence transformation maps $\triangle ABC$ onto $\triangle XYZ$? Explain.